The Deutsch-Jozsa Algorithm

Computational Basis

$$|0\rangle \leftrightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle \leftrightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Linear

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \leftrightarrow \alpha|0\rangle + \beta|1\rangle$$

Kronecker Product

$$|01\rangle \leftrightarrow |0\rangle \otimes |1\rangle \leftrightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \leftrightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

All Binary Functions: One Input

x_0	f_0	f_1	f_2	f_3
0	0	1	0	1
1	0	0	1	1

All Binary Functions: Two Inputs

x_0	x_1	f_0	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9	f_{10}	f_{11}	f_{12}	f_{13}	f_{14}	f_{15}
0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	0	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Number of Balanced Functions

$$\binom{2^n}{2^{n-1}}$$

Classical Algorithm

x_0	x_1	x_2	$f(x_0, x_1, x_2)$
0	0	0	1
0	0	1	?
0	1	0	?
0	1	1	1
1	0	0	?
1	0	1	1
1	1	0	1
1	1	1	?

Quantum Not Gate X

$$|0\rangle - \boxed{\mathrm{X}} - |1\rangle \qquad |1\rangle - \boxed{\mathrm{X}} - |0\rangle$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \leftrightarrow \quad \mathrm{X}|0\rangle = |1\rangle$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \leftrightarrow \quad \mathrm{X}|1\rangle = |0\rangle$$

Hadamard Gate H

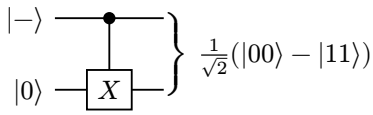
$$|0\rangle - \boxed{\mathrm{H}} - |+\rangle \qquad |1\rangle - \boxed{\mathrm{H}} - |-\rangle$$

$$|+\rangle - \boxed{\mathrm{H}} - |0\rangle \qquad |-\rangle - \boxed{\mathrm{H}} - |1\rangle$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \leftrightarrow \quad \mathrm{H}|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \leftrightarrow \quad \mathrm{H}|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Controlled X



$$\frac{1}{\sqrt{2}}\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}\begin{bmatrix}1\\0\\-1\\0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\-1\end{bmatrix}$$

$$\mathrm{CX}|-0\rangle = \mathrm{CX}\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

Deutsch Algorithm Circuit

Initialize

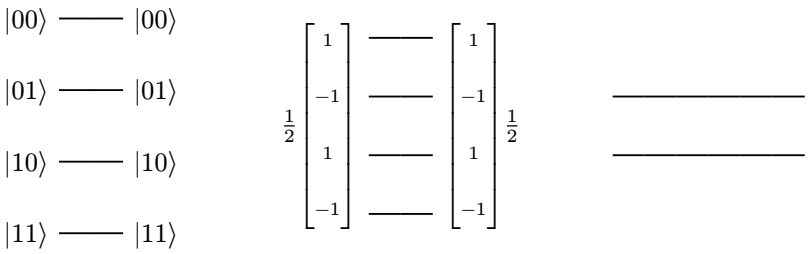
$$(\mathrm{H} \otimes \mathrm{H})|01\rangle = |+-\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$\begin{bmatrix}1&1&1&1\\1&-1&1&-1\\1&1&-1&-1\\1&-1&-1&1\end{bmatrix}\begin{bmatrix}0\\1\\0\\0\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix}$$

Oracles

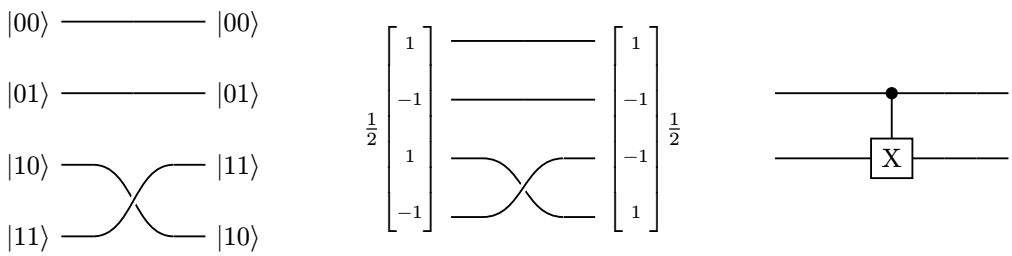
Constant 0

$\lvert xy\rangle$	x	y	$f(x)$	$y \oplus f(x)$	$\lvert x(y \oplus f(x))\rangle$
$\lvert 00\rangle$	0	0	0	0	$\lvert 00\rangle$
$\lvert 01\rangle$	0	1	0	1	$\lvert 01\rangle$
$\lvert 10\rangle$	1	0	0	0	$\lvert 10\rangle$
$\lvert 11\rangle$	1	1	0	1	$\lvert 11\rangle$



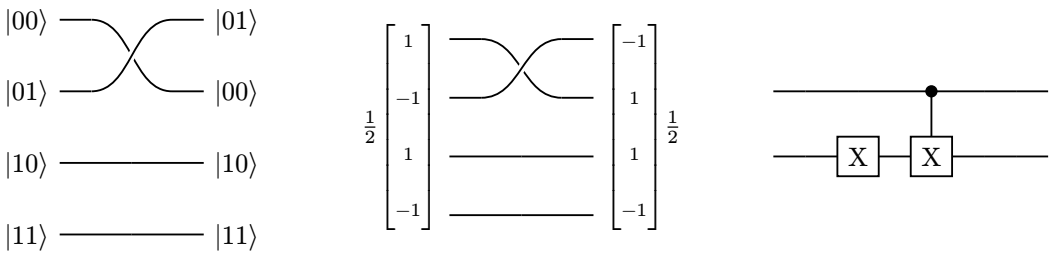
Identity

$\lvert xy\rangle$	x	y	$f(x)$	$y \oplus f(x)$	$\lvert x(y \oplus f(x))\rangle$
$\lvert 00\rangle$	0	0	0	0	$\lvert 00\rangle$
$\lvert 01\rangle$	0	1	0	1	$\lvert 01\rangle$
$\lvert 10\rangle$	1	0	1	1	$\lvert 11\rangle$
$\lvert 11\rangle$	1	1	1	0	$\lvert 10\rangle$



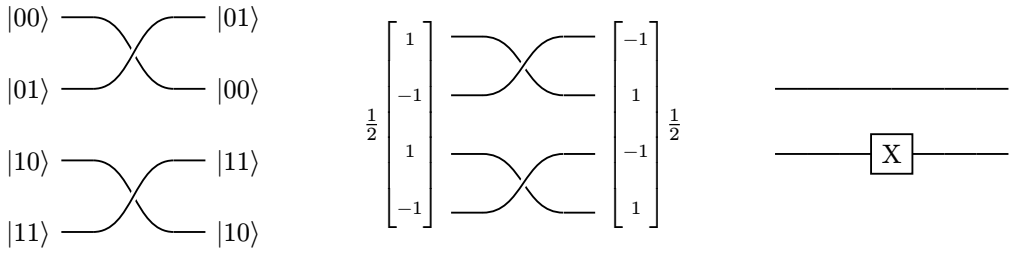
Not

$\lvert xy\rangle$	x	y	$f(x)$	$y \oplus f(x)$	$\lvert x(y \oplus f(x))\rangle$
$\lvert 00\rangle$	0	0	1	1	$\lvert 01\rangle$
$\lvert 01\rangle$	0	1	1	0	$\lvert 00\rangle$
$\lvert 10\rangle$	1	0	0	0	$\lvert 10\rangle$
$\lvert 11\rangle$	1	1	0	1	$\lvert 11\rangle$



Constant 1

$\lvert xy\rangle$	x	y	$f(x)$	$y \oplus f(x)$	$\lvert x(y \oplus f(x))\rangle$
$\lvert 00\rangle$	0	0	1	1	$\lvert 01\rangle$
$\lvert 01\rangle$	0	1	1	0	$\lvert 00\rangle$
$\lvert 10\rangle$	1	0	1	1	$\lvert 11\rangle$
$\lvert 11\rangle$	1	1	1	0	$\lvert 10\rangle$



Measurement

$$\mathrm{H} \otimes \mathrm{I} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

Constant 0

$$\mathrm{H} \otimes \mathrm{I} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 2 \\ -2 \\ 0 \\ 0 \end{bmatrix}$$

Identity

$$\mathrm{H} \otimes \mathrm{I} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 2 \\ -2 \end{bmatrix}$$

Not

$$\mathrm{H} \otimes \mathrm{I} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ -2 \\ 2 \end{bmatrix}$$

Constant 1

$$\mathrm{H} \otimes \mathrm{I} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} -2 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

First Qubit

$$\begin{aligned} 0 &\Rightarrow \text{Constant} \\ 1 &\Rightarrow \text{Balanced} \end{aligned}$$